



CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION

PHYSICS

UNIT 1 - PAPER 02

2 hours and 15 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This paper consists of **NINE** questions.
2. Section A consists of **THREE** questions. Candidates must attempt **ALL** questions in this section. Answers for this section must be written in this answer booklet.
3. Section B consists of **SIX** questions. Candidates must attempt **THREE** questions in this section, **ONE** question from **EACH** Module. Answers for this section must be written in the answer booklet provided.
4. All working **MUST** be **CLEARLY** shown.
5. The use of non-programmable calculators is permitted.

LIST OF PHYSICAL CONSTANTS

Universal gravitational constant	G	=	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Acceleration due to gravity	g	=	9.80 m s^{-2}
Radius of the Earth	R_E	=	6380 km
Mass of the Earth	M_E	=	$5.98 \times 10^{24} \text{ kg}$
Mass of the Moon	M_M	=	$7.35 \times 10^{22} \text{ kg}$
1 Atmosphere	Atm	=	$1.00 \times 10^5 \text{ N m}^{-2}$
Boltzmann's constant	k	=	$1.38 \times 10^{-23} \text{ J K}^{-1}$
Density of water		=	$1.00 \times 10^3 \text{ kg m}^{-3}$
Thermal conductivity of copper		=	$400 \text{ W m}^{-1} \text{ K}^{-1}$
Specific heat capacity of aluminium		=	$910 \text{ J kg}^{-1} \text{ K}^{-1}$
Specific heat capacity of copper		=	$387 \text{ J kg}^{-1} \text{ K}^{-1}$
Specific heat capacity of water		=	$4200 \text{ J kg}^{-1} \text{ K}^{-1}$
Specific latent heat of fusion of ice		=	$3.34 \times 10^5 \text{ J kg}^{-1}$
Specific latent heat of vaporization of water		=	$2.26 \times 10^6 \text{ J kg}^{-1}$
Avogadro's constant	N_A	=	6.02×10^{23} per mole
Molar gas constant	R	=	$8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Stefan-Boltzmann's constant	σ	=	$5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Speed of light in vacuum	c	=	$3.0 \times 10^8 \text{ m s}^{-1}$

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SECTION A

Attempt ALL questions. You MUST write in this answer booklet. You must NOT spend more than 30 minutes on this section.

1.

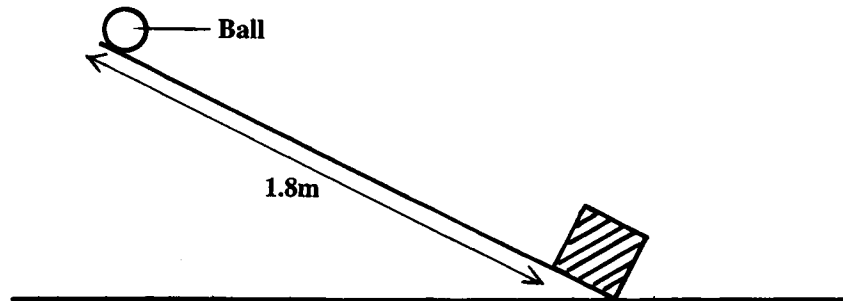


Figure 1

A ball is released on an inclined plane, Figure 1, and the distance travelled is measured at intervals as it rolls down the slope. A distance against time graph is drawn at Figure 2 on page 5.

- (a) Find the gradient of the graph when $t = 0.5$ s.

[3 marks]

- (b) Use data from the graph to find the acceleration of the ball.

[3 marks]

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- (c) Why is the gradient of the graph greater immediately before 1.2 s than immediately after 1.2 s?

[3 marks]

- (d) If further timings and displacements were taken after 1.8 s, would you expect the displacement to become zero again? Explain your answer.

[1 mark]

Total 10 marks

Distance/time graph

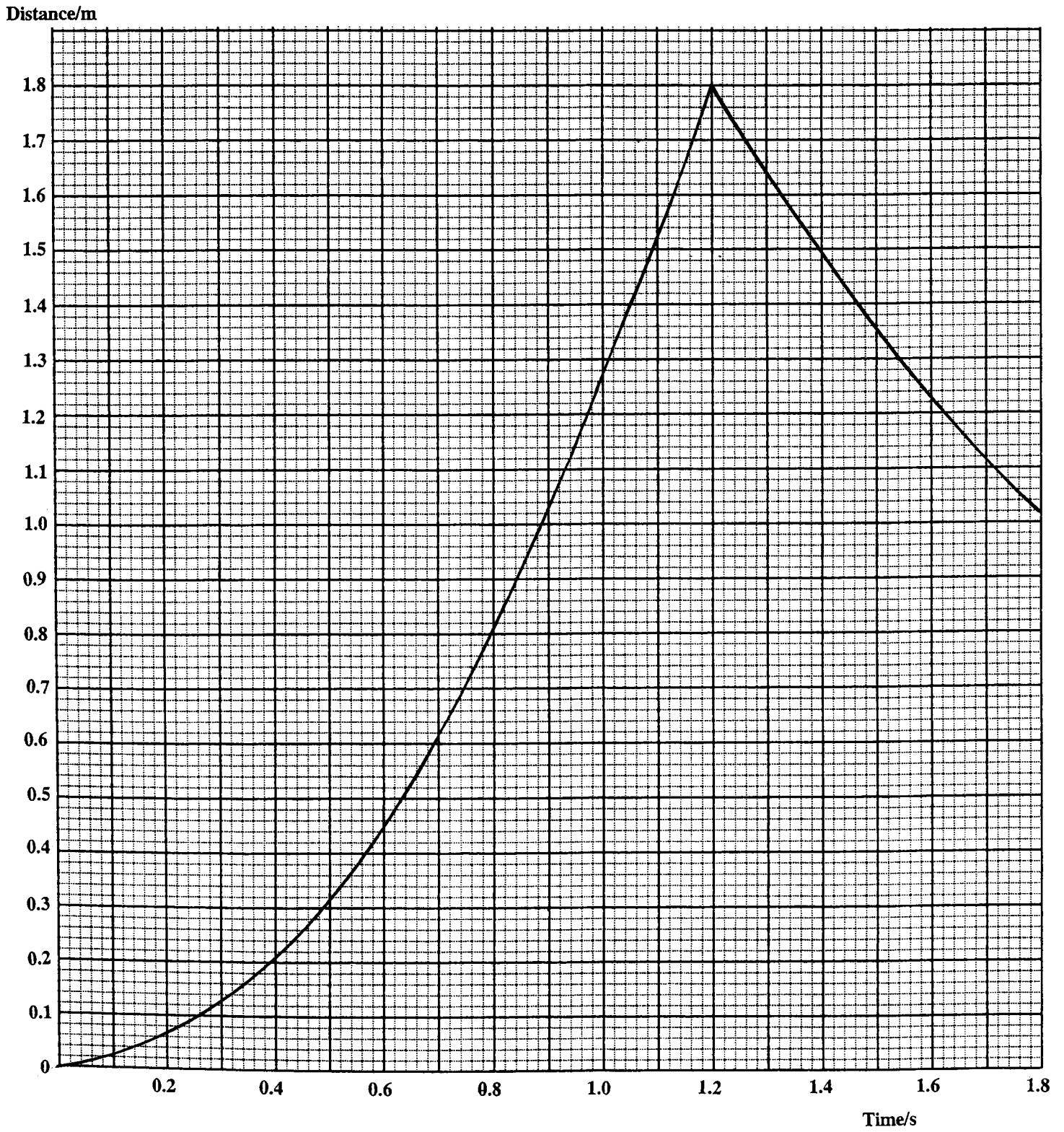


Figure 2

2. A tube with a variable length is closed at one end. A series of tuning forks is used to create resonance at the fundamental frequency for various lengths of the tube. The results obtained are shown in Table 1.

Table 1

Fundamental Frequency, f/Hz	200	250	300	400	500
Length, l/mm	406	322	264	194	153

- (a) On page 7, plot a graph of $1/f$ against l , starting the scale on the l axis at 100 mm.
[4 marks]

- (b) Use the graph to find the speed of sound during the experiment.

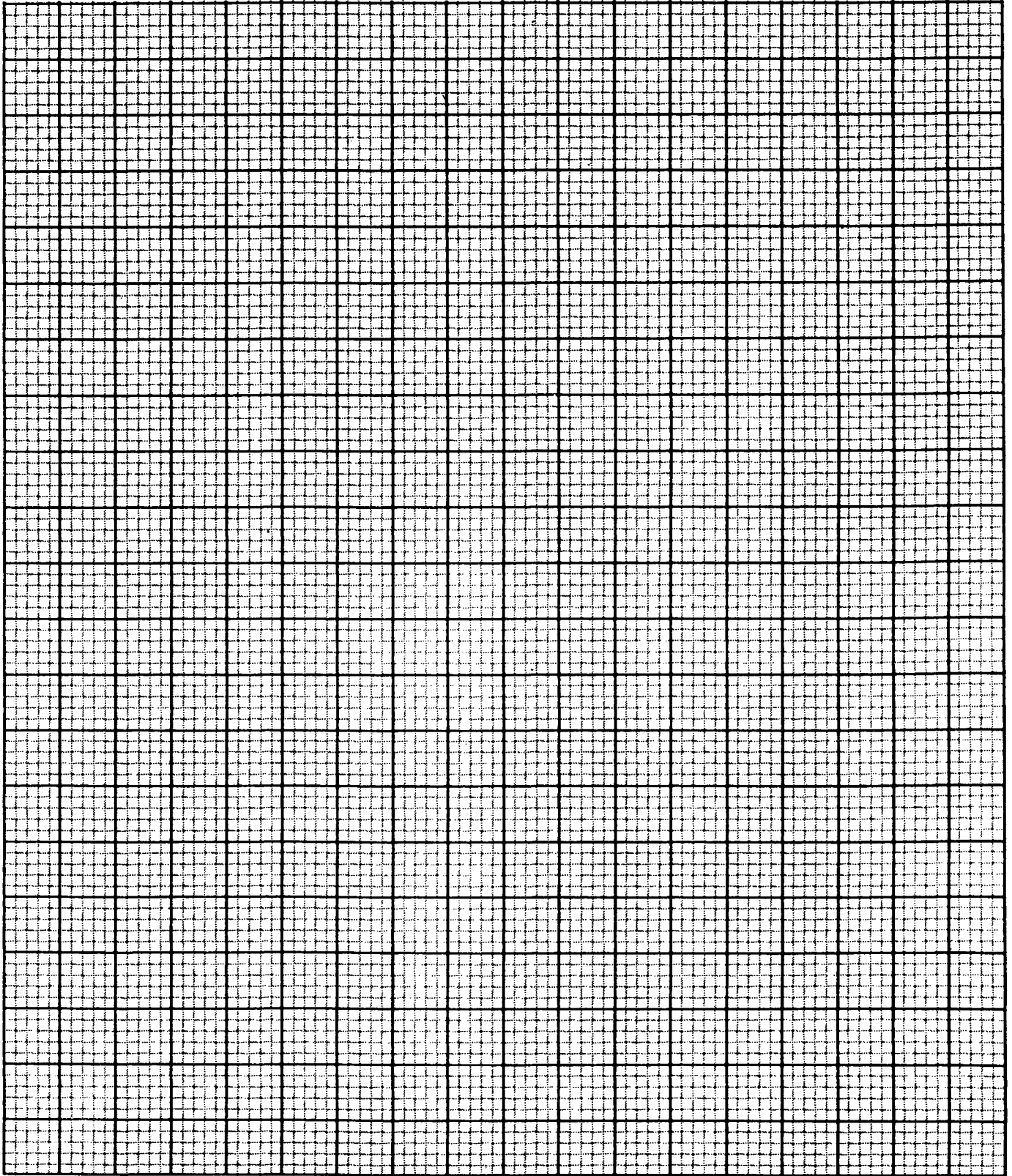
[5 marks]

- (c) Find the end connection for the tube.

[1 mark]

Total 10 marks

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3. A teacher asks her student to determine the specific heat capacity of the 2 kg aluminium block shown in Figure 3.

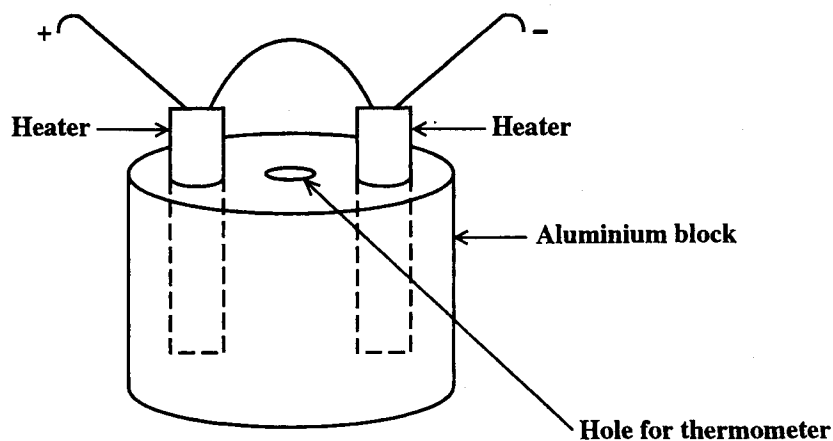


Figure 3

The student uses two heaters and each heater is rated at 100W. The student takes a series of measurements of temperature, θ , of the block at time t . The data obtained are recorded in Table 2.

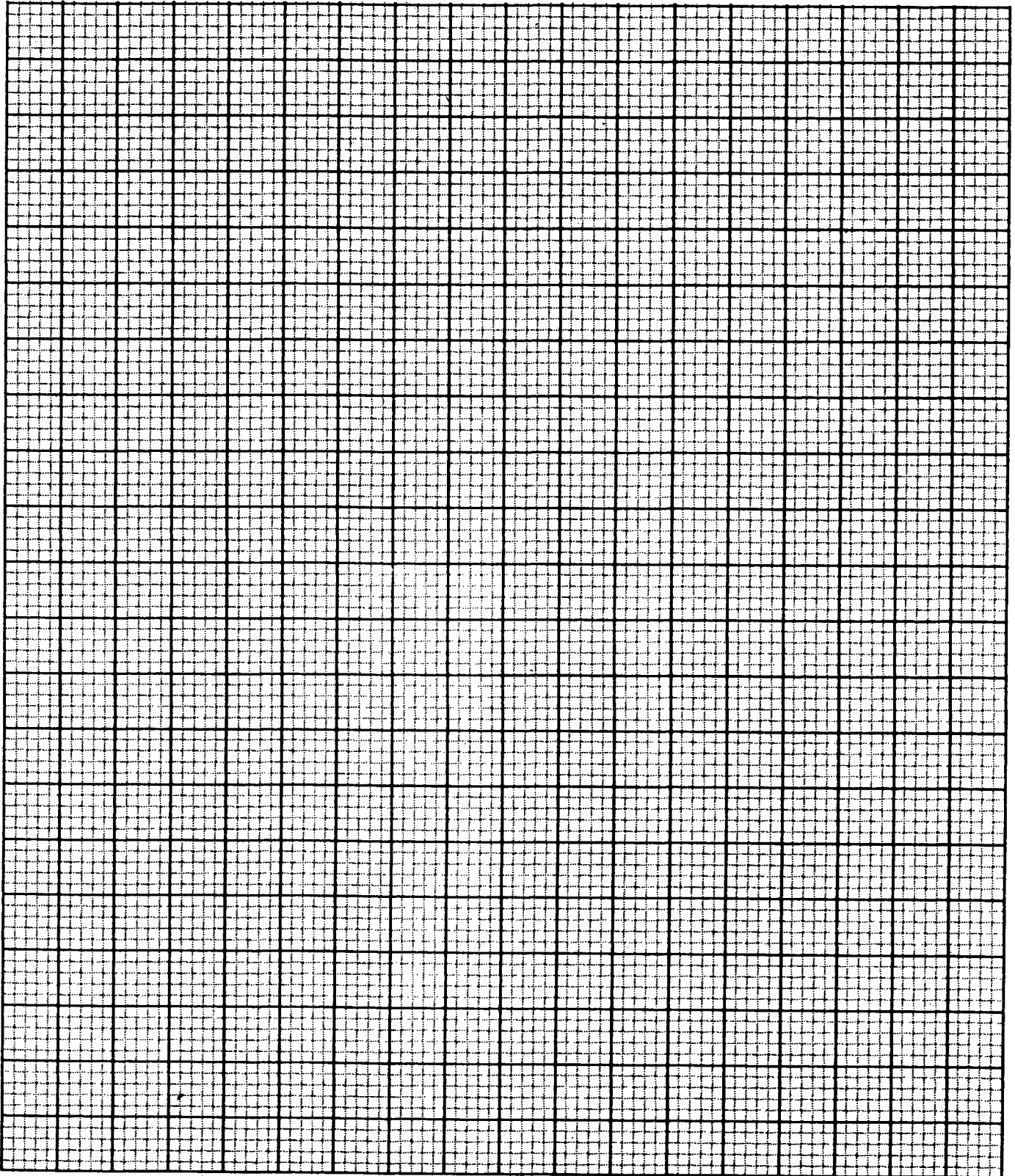
Table 2

Time, t/s	0	90	180	270	360	420
Temperature, $\theta/^\circ\text{C}$	27.1	37.2	47.0	56.8	67.0	77.3

- (a) Plot a graph of temperature versus time for the block on the grid provided on page 9. [3 marks]
- (b) From your graph determine the specific heat capacity of aluminium.

[5 marks]

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- (c) What is the heat capacity of the block?

[1 mark]

- (d) Suggest how the experiment might be improved.

[1 mark]

Total 10 marks

SECTION B

You must attempt **THREE** questions from this section. Choose **ONE** question **EACH** from Module 1, 2 and 3. You **MUST** write your answers in the answer booklet provided.

MODULE 1

Answer **EITHER** Question 4 **OR** Question 5.

4. (a) What conditions are required for a body to undergo
- (i) parabolic motion?
 - (ii) circular motion? **[4 marks]**
- (b) Explain what is meant by a 'geostationary satellite'. Show that the radius of the orbit of a geostationary satellite is independent of its mass. **[4 marks]**
- (c) A small mass of 0.60 kg is rotated at the end of a string in a horizontal circle of radius 1.20 m. The string will break if the tension exceeds 60 N. What is the **GREATEST** frequency of revolution that is possible? **[5 marks]**
- (d) The same mass of 0.60 kg is now rotated at the end of another string at a constant speed, v , in a vertical circle of radius 1.20 m. The minimum tension in the string is 2.1 N.
- (i) State and explain where the tension in the string is **MAXIMUM** and **MINIMUM**.
 - (ii) Find the speed of the mass. **[7 marks]**

Total 20 marks

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5. (a) (i) Define 'linear momentum' and state the principle of conservation of linear momentum.
- (ii) a) Distinguish between 'inelastic' collision and 'perfectly elastic' collisions.
- b) Describe how the conservation of energy applies in EACH case.
- (iii) Explain the meaning of the 'impulse of a force' and show the relation between the impulse of a force in a body and the momentum of the body. **[8 marks]**
- (b) Two trolleys are used to investigate collisions. A trolley of mass 1.60 kg is pushed and hits a second trolley, of mass 0.80 kg, which is moving in the same direction, but at a lower speed. The experiment is repeated and the results are shown below.

1st Collision

Mass	Speed before collision	Speed after collision
1.60 kg	0.70 m s ⁻¹	0.30 m s ⁻¹
0.80 kg	0.10 m s ⁻¹	0.89 m s ⁻¹

2nd Collision

Mass	Speed before collision	Speed after collision
1.60 kg	0.60 m s ⁻¹	0.37 m s ⁻¹
0.80 kg	0.10 m s ⁻¹	0.57 m s ⁻¹

- (i) Show whether or not the two sets of data are consistent with the law of conservation of momentum.
- (ii) Determine whether the collisions are elastic or inelastic.
- (iii) Why should the speeds be measured IMMEDIATELY before and after the collisions?
- (iv) The 1.60 kg trolley collided with another of the same mass, moving with the same speed, in the opposite direction.
- a) What would be the TOTAL momentum after collision?
- b) Explain your answer. **[12 marks]**

Total 20 marks

MODULE 2

Answer EITHER Question 6 OR Question 7.

6. (a) (i) Explain what is meant by 'refraction of sound waves'.
- (ii) Draw sketches to show the refraction of sound waves as the waves travel from cool air to warmer air and from warm air to cooler air.
- (iii) Hence explain why sound waves are more audible at night than in the day. **[8 marks]**
- (b) Figure 4 shows a tuning fork with a frequency of 440 Hz, held just above the top of a uniform tube containing water. The tube can excite the column of air above the water, whose level can be changed by a tap at the bottom of the tube. As the water is drained out, the sound intensity of the fork is enhanced when the air column has a length of 0.6 m and again when the air column has a height of 1 m.

Using these data, calculate a value for the speed of sound in air.

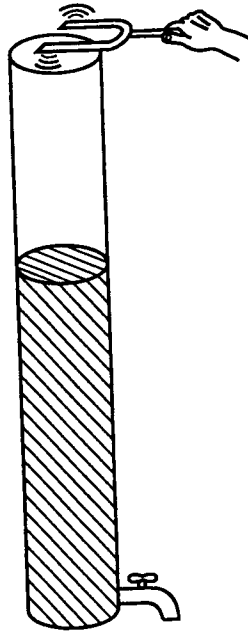


Figure 4

[4 marks]

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- (c) Figure 5 shows two loudspeakers S_1 and S_2 separated by 0.50 m. These speakers are connected to the output of an amplifier and they form sound waves with the same amplitude at a frequency of precisely 4400 Hz. The amplifier emits two waves 180° out of phase. A set of chairs is arranged in a semicircle 30.0 m from the speakers.

[Speed of sound in air = 330 m s^{-1}]

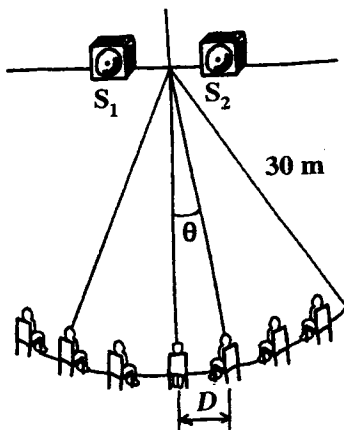


Figure 5

- (i) Calculate the amplitude of the wave at the chair on the perpendicular bisector (centre line) of the line between the speakers.
- (ii) At what MINIMUM distance, D , to the right of this central chair is there a MAXIMUM in the sound intensity? [8 marks]

Total 20 marks

7. (a) (i) Define 'simple harmonic motion' (S.H.M.) and write down an expression relating acceleration to displacement in S.H.M.
- (ii) Show that the time period of oscillation, T , of a simple pendulum of length l and mass m is given by $T = 2\pi \sqrt{\frac{l}{g}}$ where g is the acceleration due to gravity. [8 marks]
- (b) A simple pendulum has a length of 2.0 m and a mass of 0.5 kg. It is hanging from the roof of a car that is travelling at 5 m s⁻¹ and banking a corner of radius 15 m.
- (i) Calculate the time period of oscillation for the pendulum.
- (ii) If the car is now at rest, and the pendulum undergoes S.H.M, would its time period be shorter or longer than that calculated in 7 (b) (i)? Explain your answer. [6 marks]
- (c) A particle undergoes S.H.M. in which the displacement in metres is given by
- $$x = 2 \times 10^{-3} \sin 3\pi t$$
- (i) Determine the angular frequency of this oscillation.
- (ii) Find the period of this S.H.M.
- (iii) Find at time $t = 1.0$ s
- a) the displacement
- b) the velocity
- c) the acceleration. [6 marks]

Total 20 marks

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MODULE 3

Answer EITHER Question 8 OR Question 9.

8. (a) (i) State and explain the processes by which a hot body can lose heat to the surroundings.
- (ii) Explain the terms 'specific heat capacity' and 'specific latent heat' of fusion of a material. **[8 marks]**
- (b) Figure 6 shows a furnace emitting radiation through a hole of area 1.0 cm^2 .

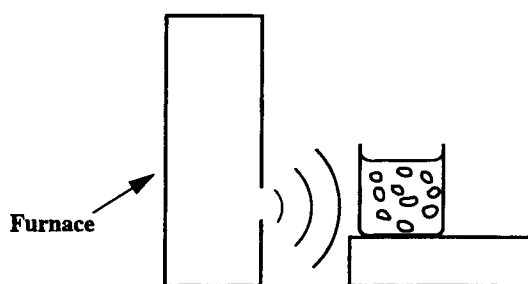


Figure 6

The radiation is absorbed by a pyrex beaker of mass 20 g containing 50 g of water and 30 g of ice in equilibrium. The temperature of the furnace is 1500°C and the specific heat capacity of pyrex is $840 \text{ J kg}^{-1} \text{ K}^{-1}$.

Calculate the time it will take for the beaker and its contents to be heated through 20°C . **[7 marks]**

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- (c) The graph of Figure 7 refers to an experiment in which a crystalline material is heated at a constant rate. The material melts at 85°C and the liquid is heated to 110°C . The specific heat capacity of the crystalline state is $430 \text{ J kg}^{-1} \text{ K}^{-1}$.

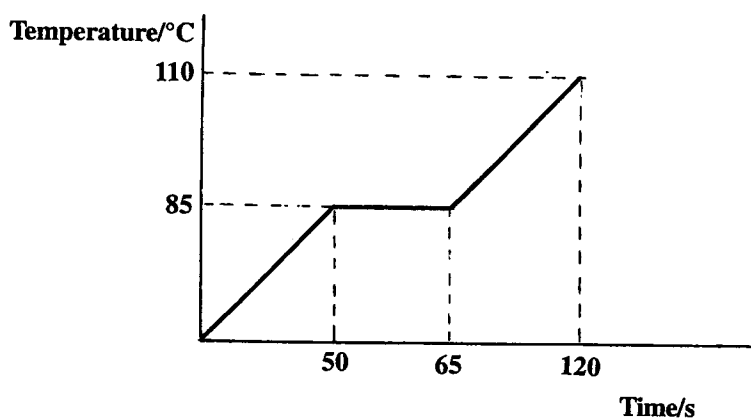


Figure 7

Calculate

- (i) the specific latent heat of fusion of the material
- (ii) the specific heat capacity of the material in the liquid phase.

[5 marks]

Total 20 marks

9. (a) (i) State the first law of thermodynamics in the form of an equation and explain the symbols used.
- (ii) Define the term 'the mole'.
- (iii) The molar heat capacity of a gas at constant pressure, c_p , differs from the molar heat capacity at constant volume, c_v . State which is the GREATER and explain why. [8 marks]
- (b) A fixed mass of an ideal monatomic gas with $c_v = \frac{3R}{2}$ undergoes two successive changes, from an initial state (P_o, V_o) to ($P_o, 3V_o$) and then to ($4P_o, 3V_o$)
- (i) Draw a graph to represent these changes.
- (ii) Show that the OVERALL change of internal energy is given by $\Delta U = \frac{33}{2} P_o V_o$.
- (iii) $P_o = 3.039 \times 10^5$ Pa and $V_o = 4 \times 10^{-3} \text{ m}^3$ calculate
- a) the work done by the gas on the surroundings
- b) the thermal energy added to the gas. [12 marks]

Total 20 marks

END OF TEST